



Operations Research

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Solution to the Training Exercises, Sheet 3

Training Exercises

Exercise T3.1. *Revised Simplex*

Problem

Solve the following linear optimization problem using the revised simplex method.

$$\begin{aligned} \text{Max} \quad & 2x_1 + x_2 + 3x_3 + x_4 + 2x_5 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 + x_5 \leq 100 \\ & x_2 + x_3 + x_4 + x_5 \leq 80 \\ & x_1 + x_3 + x_4 \leq 50 \\ & x_1, \dots, x_5 \geq 0 \end{aligned}$$

Solution

First, we transform the LP into the canonical form by introducing slack variables $x_6, x_7, x_8 \geq 0$:

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 100 \\ 80 \\ 50 \end{pmatrix} \quad c = (2, 1, 3, 1, 2, 0, 0, 0)^T$$

Iteration 1 Basic variables: $x_B = (x_6, x_7, x_8)^T$, Nonbasic variables: $x_N = (x_1, x_2, x_3, x_4, x_5)^T$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad N = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix} \quad c_B = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad c_N = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \\ 2 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad N' = B^{-1}N = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$$c'_N = -(c_N - N'^T c_B) = (-2, -1, -3, -1, -2)^T, \quad x_B = b' = B^{-1}b = (100, 80, 50)^T$$

We choose the x_3 - column as pivot column and compute the quotients of the right-hand-side (b_j) with those coefficients of the pivot column that are positive. We get 100 for the first row, 80 for the second and 50 for the third. Thus, we choose the third row as our pivot row and make the basis-exchange $x_8 \leftrightarrow x_3$.

Iteration 2 $x_B = (x_3, x_6, x_7)^\tau$, $x_N = (x_1, x_2, x_4, x_5, x_8)^\tau$

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad N = \begin{pmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix} \quad c_B = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \quad c_N = \begin{pmatrix} 2 \\ 1 \\ 2 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \quad N' = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 2 & -1 & 1 & -1 \\ -1 & 1 & 0 & 1 & -1 \end{pmatrix} \quad c'_N = \begin{pmatrix} 1 \\ -1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \quad x_B = b' = \begin{pmatrix} 50 \\ 50 \\ 30 \end{pmatrix}$$

Basis-exchange: $x_7 \leftrightarrow x_5$

Iteration 3 $x_B = (x_3, x_5, x_6)^\tau$, $x_N = (x_1, x_2, x_4, x_7, x_8)^\tau$

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad N = \begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix} \quad c_B = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \quad c_N = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{pmatrix} \quad N' = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & 0 & 1 & -1 \\ 1 & 1 & -1 & -1 & 0 \end{pmatrix} \quad c'_N = \begin{pmatrix} -1 \\ 1 \\ 2 \\ 2 \\ 1 \end{pmatrix} \quad x_B = b' = \begin{pmatrix} 50 \\ 30 \\ 20 \end{pmatrix}$$

Basis-exchange: $x_6 \leftrightarrow x_1$

Iteration 4 $x_B = (x_1, x_3, x_5)^\tau$, $x_N = (x_2, x_4, x_6, x_7, x_8)^\tau$

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad N = \begin{pmatrix} 2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \quad c_B = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad c_N = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \quad N' = \begin{pmatrix} 1 & -1 & 1 & -1 & 0 \\ -1 & 2 & -1 & 1 & 1 \\ 2 & -1 & 1 & 0 & -1 \end{pmatrix} \quad c'_N = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad b' = \begin{pmatrix} 20 \\ 30 \\ 50 \end{pmatrix}$$

The termination criterion is now fulfilled (only positive entries in c'_N), so $x = (20, 0, 30, 0, 50, 0, 0, 0)^\tau$ is an optimal solution with $Z = c_B^\tau b' = 230$.

Exercise T3.2. Lemonade Production

Problem

The lemonade factory *LiMo-United (LMU)* produces two different lemonades: *AquaMarin* and *BubbleBurst*. Both lemonades use *caffeine*, *oranges* and *sugar* as raw materials. The following table lists the available capacities of the raw materials in kilograms as well as the required quantities of the raw materials (also in kilograms) to produce 10 liters of Aquamarine or BubbleBurst.

Raw material	AquaMarin	BubbleBurst	Capacity
Caffeine	1	2	20
Oranges	2	2	30
Sugar	1	2	25
Sales price per 10 liters	12€	18€	
Costs per 10 liters	10€	15€	

Employee Mattea has already modeled the above production problem as a linear program (P) and determined the optimal table using the simplex algorithm.

$$\begin{aligned} (P) \quad & \text{Max} \quad 2x_1 + 3x_2 \\ & \text{s.t.} \quad x_1 + 2x_2 + x_3 = 20 \\ & \quad \quad 2x_1 + 2x_2 + x_4 = 30 \\ & \quad \quad x_1 + 2x_2 + x_5 = 25 \\ & \quad \quad x \in \mathbb{R}_{\geq 0}^5 \end{aligned}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	0	1	0.5	0	35
x_2	0	1	1	-0.5	0	5
x_1	1	0	-1	1	0	10
x_5	0	0	-1	0	1	5

For the following subtasks it is assumed that LMU always acts in a profit-maximizing manner.

- Explain the meaning of the objective function, the constraints and the decision variables of (P).
- How many liters of AquaMarin and BubbleBurst must be produced according to the optimal production plan?
- LMU has the option to buy another 1kg of oranges. What is the maximum price that LMU should pay for this?
- Another company was unable to obtain enough sugar due to supply shortages. It therefore proposes to buy 4kg of sugar from LMU for a total of 7€. Should LMU accept this proposal?
- Calculate the upper and lower limits for the sales price of 10L AquaMarin within which the optimal basis does not change. Briefly explain (without values) how the optimal

production plan and resource stocks change if the sales price exceeds the upper limit.

- f) Due to supply shortages, the availability of oranges is unfortunately significantly worse this year, which is why LMU is considering reducing the stock of oranges in the future.
- (i) What is the minimum number of kilograms of oranges required to continue to make it economically viable to produce both AquaMarin and BubbleBurst?
 - (ii) How do the shadow prices and reduced costs change if the amount of oranges falls below this value?
- g) Mattea proposes to produce a new drink. To produce 10 liters of *CitroCider* you need 1kg of caffeine, 1kg of oranges and 3kg of sugar. If the production costs per 10 liters are 8€ how high must the selling price of 10 liters of CitroCider be set for production to make economic sense? *Note: You can assume that all shadow prices are still valid for deviations of up to 1kg.*

Solution

- a) x_1 Amount of AquaMarin produced in multiples of 10 liters
 x_2 Amount of BubbleBurst produced in multiples of 10 liters
 x_3, x_4, x_5 Slack variables for the capacity of caffeine, oranges and sugar
Obj Profit maximization
Res1 : Do not use more than 20kg of caffeine for production
Res2 : Do not use more than 30kg of oranges for production
Res3 : Do not use more than 25kg of sugar for production
 $x_1, x_2 \geq 0$ No negative production quantities allowed, otherwise arbitrarily real.
 $x_3, x_4, x_5 \geq 0$ Slack variables must always be positive and real be.
- b) According to the optimal production plan, 100 liters of AquaMarin and 50 liters of BubbleBurst must be produced.
- c) The shadow price for an additional kilogram of oranges is 0.5€. In this respect, LMU should pay a maximum of 50 cents for an additional kilogram of oranges.
- d) The variable x_5 is the slack variable for the sugar stock. In the optimal table, you can see that 5kg of sugar are currently not being used, since $x_5 = 5$. Without changing its optimal production plan, LMU can therefore sell 4kg of sugar and thus increase its profit.
- e) We add Δ to the price of x_1 in the objective, which leads to the following in the final tableau:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	$-\Delta$	0	1	0.5	0	35
Z	0	0	$1 - \Delta$	$0.5 + \Delta$	0	$35 + 10\Delta$

The basis does not change, if all objective coefficients satisfy ≥ 0 . So $1 - \Delta \geq 0$ and $0.5 + \Delta \geq 0$. This gives Δ : $-0.5 \leq \Delta \leq 1$ and thus $p(x_1) \in [11.5, 13]$. If the price of AquaMarin rises above 13€ the coefficient of x_3 becomes negative, x_3 enters the base with value 5 and throws out x_2 . That is, LMU would stop producing BubbleBurst and instead only produce Aquamarine (150L). Oranges remain a scarce commodity, while caffeine (5kg) and sugar (10kg) would remain.

- f) (i) The slack variable for oranges is x_4 . If we increase the stock of oranges by Δ , we have to add Δ times the x_4 column to the right-hand side. Here we have to ensure that the table remains valid and that both products continue to be produced. We thus obtain the following conditions: $5 - 0.5\Delta > 0$ and $10 + \Delta > 0$, i.e. $\Delta \in (-10, 10)$. Thus, more than 20kg of oranges must be available for both drinks to continue to be produced.
- (ii) For $\Delta = -10$, the right-hand side of the second constraint is 0, and thus x_1 is the exiting base variable. We add t times the second constraint to the objective function and evaluate which variable enters the basis:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	t	0	$1 - t$	$0.5 + t$	0	30
x_1	1	0	-1	1	0	0
$Z(t = 1)$	1	0	0	1.5	0	30

Thus, $t \in [0, 1]$ would have to remain so that the tableau remains optimal. For $t \geq 1$, x_3 enters the base and the shadow prices and reduced costs change as shown in the table above.

- g) Since according to the optimal table there are still 5kg of sugar left, the shadow price for 3kg of sugar is 0€. According to the note, we can assume that the shadow prices of caffeine and oranges remain valid for deviations of up to 1kg. For oranges, we have even explicitly calculated the validity of the shadow price. The imputed costs using the shadow prices are thus: $1 \cdot 1 + 1 \cdot 0.5 + 3 \cdot 0 = 1.5$ €. Thus, 10 liters of CitroCider should be sold for at least $1.5 + 8 = 9.5$ € for production to be economically viable.

Exercise T3.3. Sensitivity

Problem

Consider the following linear program and the corresponding optimal tableau:

$$\begin{aligned}
 (P) \quad & \text{Max } -2x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + 3x_2 \leq 7 \\
 & x_1 + \frac{1}{2}x_2 \geq 1 \\
 & 2x_1 + 2x_2 \leq 4 \\
 & x \geq 0
 \end{aligned}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	5	0	0	0	$3/2$	6
x_3	-2	0	1	0	$-3/2$	1
x_2	1	1	0	0	$1/2$	2
x_4	$-1/2$	0	0	1	$1/4$	0

- Convert (P) to standard form (P_{SF}) . Is (P_{SF}) already in canonical form? Explain.
- Read an optimal solution of (P) from the given tableau.
- Is the solution of (P) unique? Explain.
- In which range can the objective function coefficient of the variable x_1 fluctuate without the optimal basis changing?
- In which range can the objective function coefficient of the variable x_2 fluctuate without the optimal basis changing?
- Determine the limits of all constraints in which the optimal basis does not change.

Solution

$$\begin{aligned}
 (P_{SF}) \quad & \underset{\text{s.t.}}{\text{Max}} \quad -2x_1 + 3x_2 \\
 \text{a)} \quad & x_1 + 3x_2 + x_3 = 7 \\
 & x_1 + 1/2x_2 - x_4 = 1 \\
 & 2x_1 + 2x_2 + x_5 = 4 \\
 & x \geq 0
 \end{aligned}$$

No, because no BV exists for the 2nd constraint.

b) $x^* = (0, 2)^T$

- c) Yes, it is. Although there is a degenerate corner and one could just as well set x_1 or x_5 as the base variable, these would still have the value 0, just like x_4 , i.e. the optimal point would not change.
- d) The objective function changes to $\max z = (-2 + \Delta)x_1 + 3x_2$. Since pivot transformations only perform addition and subtraction with the objective function coefficients $-2 + \Delta$, the following optimal table is obtained:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	$5 - \Delta$	0	0	0	$3/2$	6

So that x_1 is not included in the basis, the following must apply: $5 - \Delta \geq 0$, i.e. $\Delta \leq 5$. The objective function coefficient of x_1 can fluctuate in the range $(-\infty, 3]$ without the optimal basis changing.

- e) The objective function changes becomes $\max z = -2x_1 + (3 + \Delta)x_2$. Since pivot transformations only perform addition and subtraction with the objective function coefficients $3 + \Delta$, the following optimal table is obtained:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	5	$-\Delta$	0	0	$3/2$	6
x_3	-2	0	1	0	$-3/2$	1
x_2	1	1	0	0	$1/2$	2
x_4	$-1/2$	0	0	1	$1/4$	0

Since the objective function coefficient of a base variable is different from zero, the tableau is no longer optimal. After a further iteration step, the following new objective function is obtained:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	$5 + \Delta$	0	0	0	$3/2 + 1/2\Delta$	$6 + 2\Delta$

To ensure that the optimal basis does not change, the following must apply:

$$\begin{aligned}
 5 + \Delta &\geq 0 && \Leftrightarrow \Delta \geq -5 \\
 3/2 + 1/2\Delta &\geq 0 && \Leftrightarrow \Delta \geq -3 \\
 &\Rightarrow -3 \leq \Delta \leq \infty
 \end{aligned}$$

The optimal basis does not change if the objective function coefficient of x_2 lies in the interval $[0, \infty)$.

- f) (i) The slack variable for the first constraint is x_3 . We therefore add Δ_I times the x_3 column to the right-hand side of our final table and obtain: $1 + \Delta_I \geq 0$ i.e. $\Delta_I \geq -1$.
- (ii) The slack variable for the second constraint is x_4 . Note: x_4 was introduced as $-x_4$ in the problem. We therefore have to subtract the x_4 column in the final table from the right-hand side and obtain: $-\Delta_{II} \geq 0$, i.e. $\Delta_{II} \leq 0$.
- (iii) The slack variable for the third constraint is x_5 . We therefore add Δ_{III} times the x_5 column to the right-hand side of our final table and obtain: $1 - \frac{3}{2}\Delta_{III} \geq 0$, $2 + \frac{1}{2}\Delta_{III} \geq 0$ and $0 + \frac{1}{4}\Delta_{III} \geq 0$; i.e. $0 \leq \Delta_{III} \leq \frac{2}{3}$.

Self-study Exercises

Exercise S3.1. *Modeling*

Problem

Reformulate the following optimization problem into a linear optimization problem in normal form.

$$\begin{array}{ll}\text{Min} & \max\{x_1, x_4\} \\ \text{s.t.} & \\ & |-x_1 + x_2 + x_3 + x_4| \leq 10 \\ & \frac{x_2 - x_4}{x_1 + x_3 + 1} \leq 4 \\ & \max\{x_1, x_2\} \leq \min\{x_3, x_4\} \\ & x_1, x_2, x_3, x_4 \geq 0\end{array}$$

Solution

The optimization problem can be formulated as an LP in the following way:

For the objective function, we need a new variable $t \in \mathbb{R}_{\geq 0}$. The new objective function is $\min t$. In addition, we need two constraints $x_1 \leq t$, $x_4 \leq t$.

From $|-x_1 + x_2 + x_3 + x_4| \leq 10$ we get $-10 \leq -x_1 + x_2 + x_3 + x_4 \leq 10$.

The constraint $\max\{x_1, x_2\} \leq \min\{x_3, x_4\}$ can be replaced by four linear constraints: $x_1 \leq x_3$, $x_1 \leq x_4$, $x_2 \leq x_3$, $x_2 \leq x_4$.

The fractional constraint becomes $x_2 - x_4 \leq 4(x_1 + x_3 + 1)$.

LP in normal form:

$$\begin{array}{ll}\text{Min} & t \\ \text{s.t.} & \\ & -x_1 + t \geq 0 \\ & -x_4 + t \geq 0 \\ & x_1 - x_2 - x_3 - x_4 \geq -10 \\ & -x_1 + x_2 + x_3 + x_4 \geq -10 \\ & -x_2 + x_4 \geq 0 \\ & -x_1 + x_4 \geq 0 \\ & -x_2 + x_3 \geq 0 \\ & -x_1 + x_3 \geq 0 \\ & 4x_1 - x_2 + 4x_3 + x_4 \geq -4 \\ & x_1, x_2, x_3, x_4, t \geq 0\end{array}$$

Exercise S3.2. *Revised Simplex*

Problem

Consider the following linear optimization problem:

$$\begin{aligned}
& \text{Max } 4x_1 + 2x_2 + 3x_3 + 5x_4 \\
& \text{s.t.} \\
& \quad x_1 + x_3 + x_5 = 12 \\
& \quad 3x_1 + x_2 + 2x_3 + x_4 + x_6 = 24 \\
& \quad 2x_1 + 2x_2 + 3x_3 + 2x_4 + x_7 = 18 \\
& \quad x \geq 0
\end{aligned}$$

Perform one step of the revised simplex algorithm, starting from the initial basis $\{x_4, x_5, x_6\}$. Interpret the result. Are there any further steps necessary?

Solution

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 3 & 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 3 & 2 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 12 \\ 24 \\ 18 \end{pmatrix} \quad c = (4, 2, 3, 5, 0, 0, 0)^\tau$$

Iteration 1 Basic variables: $x_B = (x_4, x_5, x_6)^\tau$, Non-basic variables: $x_N = (x_1, x_2, x_3, x_7)^\tau$

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix} \quad N = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 \\ 2 & 2 & 3 & 1 \end{pmatrix} \quad c_B = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} \quad c_N = \begin{pmatrix} 4 \\ 2 \\ 3 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & 0 & 1/2 \\ 1 & 0 & 0 \\ 0 & 1 & -1/2 \end{pmatrix} \quad N' = B^{-1}N = \begin{pmatrix} 1 & 1 & 3/2 & 1/2 \\ 1 & 0 & 1 & 0 \\ 2 & 0 & 1/2 & -1/2 \end{pmatrix}$$

$$c'_N = -(c_N - N'^T c_B) = (1, 3, 9/2, 5/2)^\tau, \quad x_B = b' = B^{-1}b = (9, 12, 15)^\tau$$

Stopping rule is satisfied (only positive entries in c'_N), thus $x = (0, 0, 0, 9, 12, 15, 0)$ with $Z = c_B^T b' = 45$ is an optimal solution of the problem and no further steps are needed.

Exercise S3.3. Simplex - Matrix notation

Problem

Below, you are given a general LP with parameters A , an $m \times n$ matrix, c and b , vectors of sizes n and m . Note that $\text{rank}(A) = \text{rank}(A \mid b) = m < n$.

$$\begin{aligned}
& \max_{x \in \mathbb{R}^n} c^T x \\
& \text{s.t. } Ax = b \\
& \quad x \geq 0.
\end{aligned}$$

The following simplex tableau solves the above LP, where I is the identity matrix of size $m \times m$.

Base	Coefficients		Res.
	x_B	x_N	
Z	0	$-(c_N - N'^T c_B)^T$	$c_B^T b'$
x_B	I	$B^{-1}N$	$B^{-1}b$

- a) Explain every symbol that appears in the tableau $(x_B, x_N, Z, c_N, N', c_B, b', B, N, b)$: what it stands for, how it is written in terms of LP components (c, x, A, b) , whether it is a vector or a matrix, and its size.
- b) What do $B^{-1}b$, $c_B^T b'$, and $-(c_N - N'^T c_B)^T$ stand for in the optimal simplex tableau?
- c) Give some numerical values to the LP parameters c, A, b so that we have a problem with 2 variables that would require the two-phase method (or big M) for the simplex algorithm. Make sure that your LP is not unbounded.

Answer the following questions according to below optimal simplex tableau.

Base	Coefficients				Res.
	x_1	x_2	x_3	x_4	
Z	0	0	2	2	57
x_1	1	0	$2/3$	$-1/6$	2
x_2	0	1	$-1/3$	$1/3$	5

- d) Find the corresponding LP parameter, A matrix, for the above optimal simplex tableau. Note that N (the constraint coefficient matrix of x_3, x_4) is the identity matrix of size 2×2 .
- e) Find the corresponding LP parameter, b vector, for the above optimal simplex tableau.
- f) Find the corresponding LP parameter, c vector, for the above optimal simplex tableau. Note that c_1 (the objective coefficient of x_1) is equal to 6.

Solution

- a) Z is the objective function $c^T x$.

x_B and x_N are a partition of variable x . They are vectors of sizes m and $n - m$, respectively.

c_B and c_N are the parts of parameter c that correspond to the respective coefficients of x_B and x_N . They are vectors of sizes m and $n - m$, respectively.

B and N are the parts of parameter A that correspond to the respective coefficients of x_B and x_N . They are matrices of sizes $m \times m$ and $m \times (n - m)$, respectively.

b' is equal to $B^{-1}b$ where b is the right-hand side value in the given LP. It is a vector of size m .

N' is equal to $B^{-1}N$, which are already explained. It is a matrix of size $m \times (n - m)$.

- b) $B^{-1}b$ corresponds to the optimal values of the basic variables.

$c_B^T b'$ corresponds to the optimal objective value.

$-(c_N - N'^T c_B)^T$ corresponds to the shadow prices.

- c) The below LP does not have variables that can act as a basis at the start of the simplex algorithm, and the point $(0, 0)$ is not feasible. So, we need to use the two-phase method (or big M) to find a feasible basis as a starting point. It can be shown that the below

LP is not unbounded.

$$\begin{aligned} \max \quad & -x_1 - x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \\ & x_1 + x_2 \geq 1 \\ & x_1, x_2 \geq 0. \end{aligned}$$

In matrix notation, it can be written as follows.

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{R}_+^2, \end{aligned}$$

where $c = (-1 \ -1)^T$, $b^T = (4 \ -1)$, and

$$A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

d) As N is the identity matrix, we find that:

$$B^{-1} = \begin{bmatrix} 2/3 & -1/6 \\ -1/3 & 1/3 \end{bmatrix}$$

Taking its inverse gives:

$$B = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$$

Then, as the matrix N is already given as the identity matrix, we can write A as follows:

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix}$$

e) As the right hand side is equal to $B^{-1}b = [2, 5]^T$, we solve the following to find b :

$$\begin{bmatrix} 2/3 & -1/6 \\ -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

That gives:

$$\begin{aligned} b_1 2/3 - b_2 1/6 &= 2 \\ -b_1 1/3 + b_2 1/3 &= 5 \end{aligned}$$

Solving these two equalities gives $b = [9, 24]^T$.

f) We can make use of $c_B^T b' = 57$, and the given value $c_1 = 6$. As $b' = B^{-1}b = [2, 5]^T$, we have:

$$2 * 6 + 5 * c_2 = 57 \rightarrow c_2 = 9.$$

Finally, we need to find c_N . We can use the equality $-(c_N - N^T c_B)^T = [2, 2]$. We first find $N^T c_B$:

$$N^T c_B = \begin{bmatrix} 2/3 & -1/3 \\ -1/6 & 1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Then, we have $-c_N^T + [1, 2] = [2, 2]$. Hence, $c_N^T = [-1, 0]$.

Exercise S3.4.

Problem

Solve the following linear optimization problem using the revised simplex algorithm (in matrix notation):

$$\begin{array}{ll}\text{Max} & 3x_1 + 5x_2 \\ \text{s.t.} & \\ & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0\end{array}$$

Solution

We first express the LP in canonical form:

$$\begin{array}{ll}\text{Max} & 3x_1 + 5x_2 \\ \text{s.t.} & \\ & x_1 + x_3 = 4 \\ & 2x_2 + x_4 = 12 \\ & 3x_1 + 2x_2 + x_5 = 18 \\ & x_1, \dots, x_5 \geq 0\end{array}$$

In the matrix notation from the lecture:

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix} \quad c = (3 \quad 5 \quad 0 \quad 0 \quad 0)^\tau$$

Iteration 1 Basic variables: $x_B = (x_3, x_4, x_5)^\tau$, non-basic variables: $x_N = (x_1, x_2)^\tau$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad N = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \quad c_B = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad c_N = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad N' = B^{-1}N = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix}$$

$$c'_N = -(c_N - N'^\tau c_B) = \begin{pmatrix} -3 \\ -5 \end{pmatrix} \quad x_B = b' = B^{-1}b = \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$$

We choose the x_2 column as the pivot column and determine the quotients of the right-hand side (b_j) with the coefficients of the pivot column, which are positive. This gives us 6 for the second row and 9 for the third row. Thus, the second row is our pivot row, and we perform the base swap: $x_4 \leftrightarrow x_2$.

Iteration 2 $x_B = (x_2, x_3, x_5)^\tau$, $x_N = (x_1, x_4)^\tau$

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix} N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 0 \end{pmatrix} c_B = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} c_N = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix} N' = \begin{pmatrix} 0 & 1/2 \\ 1 & 0 \\ 3 & -1 \end{pmatrix} c'_N = \begin{pmatrix} -3 \\ 5/2 \end{pmatrix} x_B = b' = \begin{pmatrix} 6 \\ 4 \\ 6 \end{pmatrix}$$

Base exchange: $x_5 \leftrightarrow x_1$

Iteration 3 $x_B = (x_1, x_2, x_3)^\tau$, $x_N = (x_4, x_5)^\tau$ Note: In matrix form, we have $x_B = (x_2, x_3, x_1)^\tau$. However, the order of the variables in x_B is fundamentally irrelevant. You just have to pay attention to the correct order of the columns when creating B and N .

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 2 & 0 \end{pmatrix} N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} c_B = \begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix} c_N = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & -1/3 & 1/3 \\ 0 & 1/2 & 0 \\ 1 & 1/3 & -1/3 \end{pmatrix} N' = \begin{pmatrix} -1/3 & 1/3 \\ 1/2 & 0 \\ 1/3 & -1/3 \end{pmatrix} c'_N = \begin{pmatrix} 3/2 \\ 1 \end{pmatrix} x_B = b' = \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix}$$

The stopping rule is satisfied (only positive entries in c'_N), thus $x = (2, 6, 2, 0, 0)$ with $Z_{max} = c_B^\tau b' = 36$ is the optimal solution to the problem.

Exercise S3.5. PopCo

Problem

The snack manufacturer PopCo draws up its production plan for chocolate bars in November 2019 to maximize potential profit. It has 3 chocolate bars in its range: CrackerJoe, CrispMoe and CookieDoe. CrackerJoe requires 1 unit of each walnut, peanut and chocolate. CrispMoe requires 1 peanut and 2 chocolate. CookieDoe requires 2 walnut and 1 chocolate. This consumption, the available capacity of the materials for November and the unit price of each bar are shown in the table below. Due to machine constraints, no more than 100 pieces of CookieDoe can be produced.

Chocolate bar	CrackerJoe	CrispMoe	CookieDoe	capacity in E
Walnut	1		2	300
Peanut	1	1		400
Chocolate	1	2	1	1000
Unit price in EUR	2	3	4	

a) Set up the corresponding linear program.

Let x_1, x_2, x_3 be the variables for the quantity produced for CrackerJoe, CrispMoe and CookieDoe; s_1, s_2, s_3 are the slack variables for walnut, peanut and chocolate; s_4 is the slack variable for the CookieDoe constraint. The optimal table looks like this:

Base	Coefficients							Res.
	x_1	x_2	x_3	s_1	s_2	s_3	s_4	
z	1	0	0		3	0	4	
	1	0	0		0	0	-2	
	1	1	0		1	0	0	400
	-1	0	0		-2	1	-1	100
	0	0	1		0	0	1	

- b) Complete the optimal tableau.
- c) State the optimal production plan and the achievable profit.
- d) Calculate to what extent the price of CrispMoe can fluctuate without changing the optimal basis.
- e) What is the maximum amount that PopCo should be willing to pay for an additional unit of walnut / peanut / chocolate? Give reasons.
- f) PopCo could buy 51 unit of new peanuts for a total price of 152 EUR. Should PopCo accept the offer? Give reasons in detail. (*Note: Pay attention to the validity ranges of the shadow prices! You can assume without proof that each newly calculated shadow price is valid for deviations of up to 1 unit.*)
- g) A new bar, CrunchyFlow, uses 1 walnut and 2 peanuts. At what minimum selling price would the production of CrunchyFlow be profitable?

Solution

- a) The corresponding LP looks like this:

$$\begin{aligned}
 &\text{Max } 2x_1 + 3x_2 + 4x_3 \\
 &\text{s.t.} \\
 &\quad x_1 + \quad + 2x_3 \leq 300 \\
 &\quad x_1 + x_2 \leq 400 \\
 &\quad x_1 + 2x_2 + x_3 \leq 1000 \\
 &\quad \quad \quad x_3 \leq 100 \\
 &\quad x_1, x_2, x_3 \geq 0
 \end{aligned}$$

- b) The complete tableau looks like this:

Base	Coefficients							Res.
	x_1	x_2	x_3	s_1	s_2	s_3	s_4	
z	1	0	0	0	3	0	4	1600
s_1	1	0	0	1	0	0	-2	100
x_2	1	1	0	0	1	0	0	400
s_3	-1	0	0	0	-2	1	-1	100
x_3	0	0	1	0	0	0	1	100

- c) The complete solution can be derived from the optimal table:
 $(x_1, x_2, x_3, s_1, s_2, s_3, s_4, Z) = (0, 400, 100, 100, 0, 100, 0, 1600)$. The optimality can be recognized by the fact that all values in the objective function row are non-negative.
 CrispMoe: 400 pieces, CookieDoe: 100 pieces, no CrackerJoe; profit: 1600 EUR

- d) Let $3 + \Delta$ be the price for CrispMoe (x_2). Δ extends into the optimal table. However, we need to bring the corresponding entry back to 0 by adding the x_2 line:

Base	Coefficients							Res.
	x_1	x_2	x_3	s_1	s_2	s_3	s_4	
z	1	$-\Delta$	0	0	3	0	4	—
x_2	1	1	0	0	1	0	0	—
z	$1 + \Delta$	0	0	0	$3 + \Delta$	0	4	—

The new entries in the objective function line must remain non-negative. That is, $1 + \Delta \geq 0$ and $3 + \Delta \geq 0$. This results in $\Delta \geq -1$. The basis remains unchanged, if the price for CrispMoe is at least 2 EUR.

- e) PopCo should pay a maximum of 3 EUR for an additional unit of peanut, because that is how much profit it brings in according to the shadow price. PopCo should not pay anything for walnuts and chocolate, because they are not yet completely used up. Their shadow prices are each 0 and, according to the optimal table, there is still 100 units left of each material.
- f) We first calculate the limits for the corresponding shadow price (s_2) and add Δ times the s_2 column to the right-hand side. Since all entries must remain non-negative, we obtain the following conditions: $400 + \Delta \geq 0$ and $100 - 2\Delta \geq 0$, i.e. $\Delta \in [-400, 50]$. If the resource stock of peanuts increases by more than 50 E, the shadow price for each additional unit decreases. We determine this new price by adding the row that *causes* this limit (s_3) t times to the objective function. This means that we determine the new incoming basis variable and the resulting change in the objective function coefficients.

Base	Coefficients							Res.
	x_1	x_2	x_3	s_1	s_2	s_3	s_4	
z	1	0	0	0	3	0	4	—
s_3	-1	0	0	0	-2	1	-1	—
z	$1 - t$	0	0	0	$3 - 2t$	t	$4 - t$	—
z	0	0	0	0	1	1	3	—

All entries must be ≥ 0 . We therefore get $t \in [0, 1]$. For $t = 1$ we get the above new shadow prices and reduced costs. We see that the new shadow price for peanuts is 1 EUR and according to the hint we can assume that it is valid for at least 1 unit.

Thus, with 51 additional peanuts, PopCo makes an additional profit of $50 \cdot 3 + 1 \cdot 1 = 151 < 152$ EUR. PopCo should therefore reject the offer.

- g) According to the optimal table, 100 walnuts are still left, which means that the shadow price for 1 walnut is 0 EUR. The shadow price of peanuts is 3 EUR and, as shown in part (f), is still valid for deviations of 2 units. The profit loss from producing a CrunchyFlow is therefore $1 \cdot 0 + 2 \cdot 3 + 0 \cdot 0 = 6$. This means that the selling price must be at least 6 EUR for production to be profitable.

Exercise S3.6. Paper mill

Problem

Mertens Paper Mill converts various types of pulpwood into low, medium and high quality

newsprint. The consumption for each class, the availability of each wood species and the selling price per ton of newsprint are shown in the following table:

Pulpwood	Low	Medium	High	Availability
European beech	2	2	1	180 t
Pine	1	2	3	120 t
Spruce	1	1	2	160 t
Selling price	€ 900	€ 1000	€ 1200	

Let x_1, x_2, x_3 be the variables for the quantity of low, medium and high quality newsprint produced (in t). Employee Georg used the simplex method to determine an optimal production plan. Unfortunately, he wrote so illegibly that some entries can no longer be deciphered.

Base	Coefficients						Res.
	x_1	x_2	x_3	s_1	s_2	s_3	
z	0	200		300	300	0	
	1	0.8		0.6	-0.2	0	84
	0	0.4		-0.2	0.4	0	
	0	-0.6		-0.2	-0.6	1	

- Set up the corresponding linear program.
- Complete the optimal table above and state the optimal production plan and profit.
- What are the shadow prices and within what limits are they valid? Explain their economic significance.
- Calculate to what extent the price of low-quality newsprint can fluctuate without the optimal basis changing.
- Calculate to what extent the price of medium-quality newsprint can fluctuate without the optimal basis changing.
- Mr. Mertens could buy another ton of pine for 250 euros. Should he accept the offer? Explain your decision.
- 1t luxury newsprint uses 3t red beech, 3t pine and 3t spruce. Calculate the minimum selling price at which the production of this class of newsprint would be profitable.
- Mr. Mertens can exchange European beech for spruce on the market. For 2t spruce you get 1t European beech. How many t spruce should he exchange if the current basis in the optimal table is to be maintained?
- To simplify sales, Mr. Mertens is considering only selling newsprint in 5t increments.
 - How does this change the problem in a)?
 - Is the current optimal solution also optimal for the adapted problem? Give reasons.
 - What does the current optimal solution tell you about the achievable profit of the adapted problem?

Solution

a) The corresponding LP looks like this:

$$\begin{array}{ll}\text{Max} & 900x_1 + 1000x_2 + 1200x_3 \\ \text{s.t.} & \\ & 2x_1 + 2x_2 + x_3 \leq 180 \\ & x_1 + 2x_2 + 3x_3 \leq 120 \\ & x_1 + x_2 + 2x_3 \leq 160 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

b) The completed optimal table looks like this:

Base	Coefficients							Res.
	z	x_1	x_2	x_3	s_1	s_2	s_3	
z	1	0	200	0	300	300	0	90000
x_1	0	1	0.8	0	0.6	-0.2	0	84
x_3	0	0	0.4	1	-0.2	0.4	0	12
s_3	0	0	-0.6	0	-0.2	-0.6	1	52

optimal production plan: produce 84t of low quality newsprint, 12t of high quality newsprint, and nothing of medium quality.

Profit: 90,000 Euro

- c) The shadow prices are 300 for European beech (in $[40, 240]$), 300 for pine (in $[90, 206\frac{2}{3}]$), and 0 for spruce (in $[108, \infty)$). This means that 1 ton of European beech or pine would each bring in 300 euros more profit, while spruce has not yet been completely used up. Since 52 tons of spruce are still unused, one ton more would currently bring no profit. *Hint: bounds can be found by adding the corresponding column Δ times to the right-hand side and then finding the bounds for Δ , under the condition that all terms ≥ 0 remain – or using GUROBI*
- d) Let $900 + \Delta$ be the selling price of low-quality newsprint. Since x_1 is in the basis, the correct table must be restored by multiplying the 1st constraint by Δ and adding it to the objective function row. All entries in the objective function row must remain non-negative, i.e.: $200 + 0.8\Delta \geq 0$, $300 + 0.6\Delta \geq 0$ and $300 - 0.2\Delta \geq 0$. This gives $\Delta \geq -250$, $\Delta \geq -500$ and $\Delta \leq 1500$. This gives $\Delta \in [-250; 1500]$. So for the basis to remain unchanged, the selling price must be between $900 - 250 = 650$ and $900 + 1500 = 2400$.
- e) Let $1000 + \Delta$ be the selling price of medium-grade newsprint. The reduced cost of x_2 remains non-negative when $200 - \Delta \geq 0$, or $\Delta \leq 200$. So the basis remains unchanged as long as the selling price of medium-grade newsprint is less than $1000 + 200 = 1200$.
- f) From the shadow price for pine and its range of validity, we deduce that one tonne of pine means 300 euros extra profit. Accordingly, Mr Mertens would still make 50 euros profit by purchasing pine worth 250 euros. He should therefore accept the offer.
- g) We can use the shadow prices from the optimal table because we do not violate their limits for 1t of luxury paper. The loss of profit from producing one tonne of luxury paper is $3 \cdot 300 + 3 \cdot 300 + 3 \cdot 0 = 1800$. This means that the selling price must be at least 1800 euros for production to be profitable.

- h) Let Δ be the amount of spruce that Mr Mertens sells. So we subtract Δ times the s_3 column from the optimal table on the right-hand side, add $1/2\Delta$ times the s_1 column back on top and get $I : 84 + 3/10\Delta \geq 0$, $II : 12 - 1/10\Delta \geq 0$ and $III : 52 - \Delta - 1/10\Delta \geq 0$. Thus $\Delta \in [-280, 520/11]$. The objective function changes to $90,000 + 150\Delta$ and is thus maximized for $\Delta = 520/11$. So $520/11$ spruces should be exchanged for $260/11$ red beeches, which leads to an additional profit of $7090\frac{10}{11}$.
- i) (i) In a) additional constraints must be introduced which force the variables x_i to steps of 5, such as: $x_i = 5y_i$, $y_i \in \mathbb{Z}_{\geq 0}$, $i = 1, 2, 3$.
- (ii) The current solution would then no longer be permissible, since the amount of low and high quality newsprint produced is not divisible by 5.
- (iii) The profit determined is the upper bound for the profit of the integer problem